Chapter # 06 Work and Energy

Define work, write its mathematical form and units. Why work is scalar quantity while moment of force is a vector?

A man is pushing the truck but truck is at rest, is he doing work? Explain.

Muhmmad Tuaha was pushing a box with a force of 100 N on a flat frictionless surface. How much work he does on the box if he pushes it through a displacement of 200 m?

Define energy, write its unit? How it is different from work although they have same units?

Briefly explain the forms of energy.

What is Kinetic energy, derive its formula. List two examples of daily life of Kinetic energy.

Babar Azam hits a cover drive by giving kinetic energy of 50 J to the ball by his bat. At what speed will the ball go to the boundary if mass of the ball is 120 g?

What is Potential energy, derive its formula. List two examples of daily life of potential energy.

At School sports gala, Kamil made a record in high jump at his school. He jumped 1.5 m high. What will be his potential energy if his mass is 60 kg? From where does this potential energy come?

What is law of conservation of energy? Derive its mathematical expression if a ball is dropped from the roof of a building. Now includes the effect of friction in the expression.

A player kicks the football of mass 450 g is thrown vertically upward at a speed of 22 ms^{-1}.
(a) Neglecting air resistance how high would it reach. (b) If due to air friction the ball only rises to 20 m, what is the work done against air resistance?

Differentiate between renewable energy and non-renewable energy? Give examples of both.

What are fossil fuels? Write a general chemical equation of combustion of methane. Write any three important fossil fuels?

Briefly describe nuclear fuel. Give its example.

Briefly describe the bio-mass fuel? Give examples. What are best uses of biomass fuels?

What are hydroelectric resources of energy? How this energy is produced, briefly explain.

What are solar radiations and how do we get energy from them?

Briefly explain the following resources of energy: geothermal , wind, wave in the sea, and tides.

Define Power. Write its equation and units?

Show that 1KHW = 3.6 x10^6 J

How powerful will be a Tesla Car's engine if it does 3×10^7 J of work in 1 minute? What will be its power in hp?

What is efficiency and why do we measure efficiency devices like electric generators?

A petrol engine in-takes 400 J of fuel energy and does only 120 J of useful work. How efficient is this petrol engine?

A car is moving with a constant speed along a straight road. Is there any work done on the car?

👉 **Net work done on the car is zero.**

Explanation:

- Constant speed → **no change in kinetic energy**
- By work-energy theorem → **Net work = change in KE = 0**

👉 Although the engine is doing work to overcome **friction and air resistance**, this is balanced by opposing forces.

A car has Kinetic energy 'E_K'. By what factor its kinetic energy would change, if its velocity is doubled?

A bullet is fired from gun, bullet penetrates into sand wall and it stops. Where does its kinetic energy used?

An LED light bulb has efficiency of 80%. Does it violate conservation of energy principle?

1) If velocity is doubled, by what factor does kinetic energy change?

$$KE = \frac{1}{2}mv^2$$

👉 Since $KE \propto v^2$:
If velocity is doubled $(v \to 2v)$,

$$KE \to (2)^2 = 4 \text{ times}$$

2) Where does the kinetic energy of the bullet go?

When the bullet stops:

- Converted into **heat energy** (due to friction)
- Used in **deforming the bullet and sand**
- Produces **sound energy**

✅ **Answer:** KE is transformed into **heat, deformation, and sound**.

3) LED efficiency = 80%. Does it violate conservation of energy?

No, it does not violate the law.

👉 80% energy becomes useful light
👉 Remaining 20% is lost as **heat**

✅ **Answer:** Energy is **not lost**, only **converted into other forms**, so conservation law is satisfied.

Will we eventually rely entirely on renewable energy sources? Why or why not?
How can increasing the power of a machine impact its energy consumption?
. A perpetual engine has an efficiency equal to 1. Why it will not work?

1) Will we eventually rely entirely on renewable energy sources? Why or why not?

Possibly in the future, but not completely yet.

👉 Renewable sources (solar, wind, hydro) are **clean and sustainable**, but:

- They are **intermittent** (depend on weather)
- Require **storage and infrastructure**

Conclusion: We are moving toward renewables, but **full dependence will take time**.

2) How can increasing the power of a machine impact its energy consumption?

- Power = energy used per second
  👉 Higher power → **more energy consumed in less time**
  👉 If used longer, **total energy consumption increases**

Conclusion: Increasing power generally **increases energy usage rate**.

3) A perpetual engine has efficiency = 1. Why will it not work?

- Efficiency = 1 means **100% energy conversion** (no loss)
  👉 This is impossible because:
- Some energy is always lost as **heat, friction, etc.**
- Violates **laws of thermodynamics**

Conclusion: A perpetual engine cannot exist because **energy losses cannot be eliminated**.

(Ans: 750 J)

A boy weighing 75 N jumps up and gains 300 J of gravitational potential energy. Find the height to which the boy will rise.

Given:

- Weight $W = 75\,N$
- Potential Energy $PE = 300\,J$

Formula:

$PE = W \times h$

Calculation:

$$300 = 75 \times h$$

$$h = \frac{300}{75} = 4\,m$$

(Ans: 300 J and 15.5 m/s)

A 2.0 kg rock is dropped from 20 m tall building. What is the Kinetic and gravitational potential energy when the rock has fallen 15 m.

Given:

- Mass $m = 2.0\,kg$
- Total height = 20 m
- Fallen distance = 15 m → Remaining height = **5 m**
- $g = 9.8\,m/s^2$

1) Gravitational Potential Energy (at 5 m):

$$PE = mgh$$

$$PE = 2 \times 9.8 \times 5 = 98\,J$$

2) Kinetic Energy

Total initial energy at top:

$$E_{total} = mgh = 2 \times 9.8 \times 20 = 392\,J$$

Using conservation of energy:

$$KE = E_{total} - PE = 392 - 98 = 294\,J$$

(Ans: (a) 2250 J, (b) 900 J and (c) 22.8 %

What horsepower (hp) is required to pump up 2500 kg of water to 100 m height in 5 minutes?

Given:

- Mass $m = 2500\,kg$
- Height $h = 100\,m$
- Time $t = 5\,min = 300\,s$
- $g = 9.8\,m/s^2$

Step 1: Work done

$$W = mgh = 2500 \times 9.8 \times 100 = 2{,}450{,}000\,J$$

Step 2: Power

$$P = \frac{W}{t} = \frac{2{,}450{,}000}{300} \approx 8167\,W$$

Step 3: Convert to horsepower

$$1\,hp = 746\,W$$

$$P = \frac{8167}{746} \approx 10.95\,hp$$

Chapter # 07 Density and Temperature

Define density, write its formula and units? State the densities of solids, liquids, and gasses? (See Book)

You find a material in the shape of cube of side length 5 centimetre. Mass of this cube is 500 grams. Find the density of this material?

Given: Mass of the material = 500 g

Side length of the cube of the given material = 5 cm

Required: Density of the material $\rho = ?$

Solution: Volume of a cubic shape object is given by: $Volume = (side\ length)^3$

Putting values $Volume = (5\,cm)^3 = 125\,cm^3$

Now, we will find density of cube shaped object is:

$$Density\ of\ cube\ shaped\ object = \frac{Mass\ of\ cube}{Volume\ of\ cube}$$

Putting values $\rho = \dfrac{500\,g}{125\,cm^3}$

Therefore, $\rho = 4\,g/cm^3$

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Compare the properties of solids, liquids, and gasses? (See Book)

Why Pasma is known as fourth state of matter? What are the sources of plasma and write its three applications?

Plasma is called the fourth state because it is an **ionized gas** in which atoms lose or gain electrons, forming **free electrons and ions**.
👉 It behaves differently from solids, liquids, and gases and **conducts electricity** and responds to magnetic fields.

Sources of Plasma

- **Natural:** Sun, stars, lightning, auroras
- **Artificial:** Neon signs, fluorescent lamps, plasma TVs

Three Applications of Plasma

1. Fluorescent and neon lights
2. Plasma TVs/displays
3. Plasma cutting and welding in industry

How does temperature affect the motion of particles? Also draw graph between temperature and K.E. of particles, and Temperature vs Pressure of particles? (See Book)

What is internal energy and how it is related to temperature of a substance?

Internal Energy (Definition)

Internal energy is the **total energy stored inside a substance**, which includes:

- **Kinetic energy** of its particles (due to motion)
- **Potential energy** of its particles (due to their positions and forces between them)

Relation with Temperature

- Temperature is a measure of the **average kinetic energy of particles**.
- When temperature **increases**, particle motion increases → **kinetic energy increases** → **internal energy increases**.
- When temperature **decreases**, internal energy also **decreases**.

How can internal energy of a substance be increased without adding heat?

✅ Method: Mechanical Work (Compression)

- When a gas is **compressed**, work is done **on the gas**.
- This increases the **kinetic energy of its particles**, so **internal energy increases**.

👉 Example:

- Pumping air into a bicycle tire — the pump gets warm.
- Compressing air in a cylinder.

✅ Another Example: Friction

- Rubbing hands together increases internal energy due to **work done against friction** (no heat supplied externally).

Briefly describe diverse types of thermometers like Mercury , Gas Thermometer and its types like Constant Pressure and constant volume, also explain liquid crystal thermometers.

Types of Thermometers

1) Mercury Thermometer

- Uses **mercury** as the working liquid.
- Works on **uniform expansion of mercury with temperature**.
- Mercury is visible, does not stick to glass, and has a **high boiling point**, so it gives **accurate readings**.
- Used in clinical and laboratory thermometers.

2) Gas Thermometer

- Uses a **gas** whose pressure or volume changes with temperature.
- Very **accurate**, used as a standard thermometer.

Types:

a) Constant Volume Gas Thermometer

- Volume is kept constant.
- **Pressure changes with temperature**.

  👉 Temperature ∝ Pressure

b) Constant Pressure Gas Thermometer

- Pressure is kept constant.
- **Volume changes with temperature**.

  👉 Temperature ∝ Volume

3) Liquid Crystal Thermometer

- Uses **liquid crystals** that change color with temperature.
- Different colors represent different temperatures.
- Used in **medical strips (forehead thermometers)** and mood rings.

Differentiate between Thermistors and Thermocouples thermometers.

Thermistors	Thermocouples
Made of **semiconductor material**	Made of **two different metals joined together**
Work on **change in resistance with temperature**	Work on **Seebeck effect (voltage produced due to temperature difference)**
Very high sensitivity	**Moderate sensitivity**
Used for **small temperature range**	Used for **very wide temperature range**
More **accurate for low temperatures**	Suitable for **high temperatures**
Slower response	Faster response
Example: digital thermometers	Example: industrial temperature sensors

Briefly explain the sensitivity, range, and linearity of Thermometers.

1) Sensitivity

Sensitivity is the **ability of a thermometer to detect small changes in temperature**.
👉 It is measured by the **change in length of liquid per degree (°C)**.
👉 Higher sensitivity means **small temperature change → large rise in liquid**.

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2) Range

Range is the **minimum and maximum temperature** that a thermometer can measure.
👉 It depends on the **freezing and boiling points of the liquid used**.

3) Linearity

Linearity means that the thermometer shows **equal expansion for equal temperature rise**.
👉 A linear thermometer gives **uniform and accurate readings** throughout its scale.

How the structure of a liquid-in-glass thermometer affects its sensitivity, range, and linearity.

🔷 1) Sensitivity

Sensitivity = change in length per °C

Depends on:

- **Narrow capillary tube** → more rise → **high sensitivity**
- **Large bulb volume** → more liquid expansion → **high sensitivity**
- **Liquid with high expansion coefficient** → increases sensitivity

👉 **Conclusion:** Narrow tube + large bulb = more sensitive thermometer

🔷 2) Range

Range = temperature interval measurable

Depends on:

- **Length of capillary tube** → longer tube = **greater range**
- **Boiling & freezing points of liquid** → wider limits = **greater range**

👉 **Conclusion:** Long tube + suitable liquid = wider range

🔷 3) Linearity

Linearity = equal expansion for equal temperature rise

Depends on:

- **Uniform bore (constant diameter) of capillary**
- **Nature of liquid (should expand uniformly)**

👉 **Conclusion:** Uniform tube + suitable liquid = accurate linear scale

Why water is not used in liquid in glass thermometres?

Water is not suitable due to the following reasons:

1. **Irregular expansion**
   Water does not expand uniformly with temperature (especially between 0°C–4°C), so readings become inaccurate.
2. **Low boiling point (100°C)**
   It cannot measure high temperatures.
3. **Freezing point at 0°C**
   It freezes easily, so it cannot measure very low temperatures.
4. **Wets glass**
   Water sticks to the walls of the glass, making it difficult to read the level clearly.
5. **Low visibility**
   It is colorless, so it is hard to see without adding dye.

Why is mercury used in thermometers?

Mercury is preferred due to these important properties:

1. **Uniform (regular) expansion**
   It expands evenly with temperature → gives **accurate readings**.
2. **Clearly visible**
   It is **shiny and opaque**, so the level can be seen easily without dye.
3. **Does not wet glass**
   It does not stick to the walls of the tube → forms a **clear meniscus**.
4. **Wide temperature range**
   - Low freezing point (−39°C)
   - High boiling point (357°C)
     → Suitable for many temperatures.
5. **Good thermal conductivity**
   Responds quickly to temperature changes.

Can we increase internal energy of a substance without increasing its temperature?

Yes, it is possible.

👉 Internal energy can increase **without a rise in temperature during a change of state** (phase change).

Explanation:

- When heat is supplied during melting or boiling, the temperature remains **constant**.
- The added energy is used to **break intermolecular forces**, not to increase kinetic energy.
- This energy is called **latent heat**, and it increases the **internal energy**.

Example:

- Ice at 0°C → Water at 0°C (internal energy increases, temperature stays same)
- Water at 100°C → Steam at 100°C

Why - 273.15°C temperature is called absolute zero? Can we achieve this temperature?

Why is −273.15°C called absolute zero?

- −273.15°C (or **0 K**) is the **lowest possible temperature**.
- At this temperature, the **kinetic energy of particles becomes minimum** (ideally zero).
- Molecular motion nearly stops, so it is called **absolute zero**.

Can we achieve this temperature?

No, absolute zero cannot be exactly achieved.

👉 According to thermodynamics:

- We can **approach** 0 K very closely
- But we can **never reach it exactly**

To increase sensitivity **without changing range**, we can:

- Use a **narrow capillary tube** (greater rise for same expansion)
- Use a **liquid with higher coefficient of expansion**

👉 At the same time, adjust the **length of the scale** so the overall range remains unchanged.

3. A 70cm 10cm 30cm plastic box has mass of 2500 g. Find the density of plastic.

(Ans. 0.12 g/cm^3)

Given:

- Length = 70 cm
- Width = 10 cm
- Height = 30 cm
- Mass = 2500 g

Volume:

$$V = 70 \times 10 \times 30 = 21000\, cm^3$$

Density:

$$\rho = \frac{m}{V} = \frac{2500}{21000}$$

$$\rho \approx 0.12\, g/cm^3$$

4. Aluminum has a density of 2700 kg/m^3. Find the mass of a solid 25 cm diameter aluminum ball.

(Ans. 22 kg)

Given:

- Density = 2700 kg/m^3
- Diameter = 25 cm → Radius = 12.5 cm = 0.125 m

Volume of Sphere:

$$V = \frac{4}{3}\pi r^3$$

$$V = \frac{4}{3}\pi(0.125)^3 \approx 0.00818\, m^3$$

Mass:

$$m = \rho V = 2700 \times 0.00818$$

$$m \approx 22\, kg$$

5. A cube of iron has a side length of 10 cm. What is volume of this cube? Mass of this iron cube is half kilogram. This cube has cavity inside it, find the volume of the cavity?

(Ans. 935.9 cm^3)

Given:

- Side = 10 cm
- Mass = 0.5 kg = 500 g
- Density of iron ≈ 7.8 g/cm^3

Volume of Full Cube:

$$V_{cube} = 10^3 = 1000\,cm^3$$

Actual Volume of Iron Present:

$$V = \frac{m}{\rho} = \frac{500}{7.8} \approx 64.1\,cm^3$$

Volume of Cavity:

$$V_{cavity} = 1000 - 64.1 \approx 936\,cm^3$$